

bWAPP on Kali Linux

Installation notes and troubleshooting report

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A reusable guide based on the installation you confirmed working: Docker, PHP 7.4, MySQL 5.7, and the Debian Bullseye security archive fix.

Recorded outcome

The database image downloaded. The PHP web image initially failed with Debian security package 404 errors. After the archive repair and rebuild, you confirmed bWAPP worked.

Item	Configuration
Lab folder	~/Desktop/labs/bWAPP
Web runtime	php:7.4-apache; Bullseye in the supplied build log
Database	mysql:5.7; Compose service name db
Web mapping	127.0.0.1:8080:80
Login	bee / bug
Database persistence	Named volume db_data

Use this guide

Pages 2–5: fresh installation. Page 6: restart commands. Page 7: troubleshooting. Page 8: evidence and references.

Use the browser inside Kali. The upstream application is deliberately vulnerable; the web port is bound to loopback and MySQL is kept on the internal Docker network.

Evidence comes from your supplied log and confirmation. Exact Kali, Docker, and Compose host versions were not provided. These notes do not claim an independent connection to your machine.

Install tools and create the lab

1 / Install Docker, Compose, Git, and Python

```
sudo apt update
sudo apt install -y docker.io docker-compose git python3
sudo systemctl start docker
sudo docker compose version
```

Keep sudo on Docker commands. If your installation only supports docker-compose, use that spelling throughout. The first command refreshes Kali package lists; the later archive repair affects the image build only.

2 / Clone the upstream repository

```
mkdir -p ~/Desktop/labs
cd ~/Desktop/labs
git clone https://github.com/lmoroz/bWAPP.git
cd bWAPP
```

If the bWAPP folder already exists, enter that folder instead of cloning again. This guide uses ~/Desktop/labs/bWAPP to match the working installation.

3 / Record the checkout when needed

```
git rev-parse HEAD
```

The upstream master revision inspected for these notes was d34520de7dfad526913706e9772aae6e9aa34916. Your exact installed revision was not captured.

Requirements

Kali desktop or VM, sudo access, and network access to package repositories, GitHub, and Docker image downloads. The supplied build log used amd64. Legacy MySQL 5.7 image availability can differ on ARM systems.

Configure the web and database containers

Run the following in ~/Desktop/labs/bWAPP. It creates a separate Compose file while leaving the upstream Compose file available.

```
cat > compose.local.yml <<'EOF'
services:
  web:
    build: .
    ports:
      - "127.0.0.1:8080:80"
    depends_on:
      - db

  db:
    image: mysql:5.7
    environment:
      MYSQL_ALLOW_EMPTY_PASSWORD: "yes"
    volumes:
      - db_data:/var/lib/mysql

volumes:
  db_data:
EOF
```

How the connections work

Connection	Meaning
Kali browser → 127.0.0.1:8080	Docker forwards to Apache on web container port 80.
Web container → db:3306	MySQL is reached through the Compose network.
db_data → /var/lib/mysql	The database files persist outside the container lifecycle.

The upstream database settings already use host db, user root, and an empty password. No MySQL installation on Kali is needed.

Repair the recorded Debian 404 error

The failing build step installed iputils-ping and dnsutils. Its package index update completed, but security package downloads returned 404. Debian documents the Bullseye archive transition; see the references on page 8.

The block below applies the working repair, saves the original Dockerfile once, and skips a repeat run if the fix is already present. Copy the complete block inside the upstream bWAPP folder.

```
python3 - <<'PY'
from pathlib import Path

path = Path("Dockerfile")
text = path.read_text()
old = "RUN apt-get update && apt-get install -y iputils-ping dnsutils"
new = (
    'RUN sed -i '
    '"s|http://deb.debian.org/debian-security|'
    'https://archive.debian.org/debian-security|g" '
    '/etc/apt/sources.list && rm -rf /var/lib/apt/lists/*\n'
    'RUN apt-get -o Acquire::Check-Valid-Until=false update '
    '&& apt-get install -y iputils-ping dnsutils'
)
if "archive.debian.org/debian-security" in text and (
    "Acquire::Check-Valid-Until=false" in text
):
    print("Archive fix already present; Dockerfile unchanged.")
elif old in text:
    backup = path.with_name("Dockerfile.before-fix")
    if not backup.exists():
        backup.write_text(text)
    path.write_text(text.replace(old, new, 1))
    print("Dockerfile patched; original backup preserved.")
else:
    raise SystemExit("Dockerfile differs: review before applying this fix.")
PY
```

The archive URL and validity-date option apply inside the container build. Package signatures remain checked. Archived packages do not receive normal ongoing security updates.

Build, initialise, and verify

4 / Build the web image

```
sudo docker compose -f compose.local.yml build --no-cache web
```

Wait for a successful build before continuing. `--no-cache` reruns the build layers. The existing MySQL image does not need a rebuild.

5 / Start the services

```
sudo docker compose -f compose.local.yml up -d  
sudo docker compose -f compose.local.yml ps
```

Both web and db should be running. MySQL can need extra time to become ready even after its container starts.

6 / Create the database — first installation only

Open <http://127.0.0.1:8080/bWAPP/install.php> in Firefox inside Kali, then click here. If the database is still starting, wait about 30 seconds and refresh.

7 / Log in

Open <http://127.0.0.1:8080/bWAPP/login.php>.

```
Username: bee  
Password: bug
```

Completion checks

Confirm that the web image built, both services are running, the installer created the database, and the login page accepts the default account. Your original installation outcome was confirmed in the conversation; no extra outputs or screenshots are invented here.

Next session: keep this page handy

Start after restarting Kali

```
cd ~/Desktop/labs/bWAPP
sudo systemctl start docker
sudo docker compose -f compose.local.yml up -d
```

Then open <http://127.0.0.1:8080/bWAPP/login.php> and use bee / bug. You do not need to rebuild or rerun `install.php` for a normal restart.

Stop when finished

```
sudo docker compose -f compose.local.yml stop
```

Check status and logs

```
sudo docker compose -f compose.local.yml ps
sudo docker compose -f compose.local.yml logs --tail=60
```

Remove containers and retain the database

```
sudo docker compose -f compose.local.yml down
```

The named database volume remains. Adding `-v` removes that volume and its data; avoid it when retaining your lab.

Optional: start Docker at boot

```
sudo systemctl enable docker
```

This starts the Docker service at boot. Bring up this Compose stack with the start commands above.

Troubleshooting: match the failing stage

Symptom	Next action
Bullseye security package 404	Check the archive patch and rebuild web with --no-cache.
Docker daemon unavailable	Start Docker and check its service status.
Docker socket permission denied	Use sudo for the Docker command.
Compose file not found	Enter ~/Desktop/labs/bWAPP and check the filename.
Port 8080 occupied	Inspect the listener. Change the host mapping to 8081 if needed.
Database connection refused	Check db status and logs; wait for MySQL to become ready.
Browser connection refused	Use Kali browser; check web status, logs, and port mapping.
New main-repository 404	Check the exact failing URL and current Debian archive status.

Useful diagnostics

```
sudo systemctl status docker --no-pager
ls compose.local.yml
sudo ss -ltnp 'sport = :8080'
sudo docker compose -f compose.local.yml logs --tail=60 db
sudo docker compose -f compose.local.yml logs --tail=60 web
```

If you choose port 8081

Change the web mapping in compose.local.yml to 127.0.0.1:8081:80, run up -d, and use port 8081 in the browser URLs. Inspect an existing service before stopping it.

The repair covers the security repository in the supplied Bullseye log. If upstream files or repository locations change, inspect the new failure instead of applying unrelated fixes.

Evidence, limits, and source references

Installation record

Check	Evidence
Initial failure	Supplied terminal log: package downloads returned 404; exit 100.
Repair result	User confirmed that bWAPP worked after the archive repair.
Upstream configuration	Repository README, Dockerfile, Compose file, and settings inspected.
Reusable helper	Patch application, repeat run, backup preservation, and unexpected-input checks.
Independent fresh Docker build	Not performed while generating these documents.

References checked for these notes

Upstream repository and setup

<https://github.com/lmoroz/bWAPP/tree/master>

Upstream Dockerfile

<https://github.com/lmoroz/bWAPP/blob/master/Dockerfile>

Upstream installation / credentials

<https://github.com/lmoroz/bWAPP/blob/master/INSTALL.txt>

Installing Docker on Kali

<https://www.kali.org/docs/containers/installing-docker-on-kali/>

Debian Bullseye LTS end of life

<https://www.debian.org/News/2026/20260831>

Debian archive transition, bug 1147093

<https://bugs.debian.org/cgi-bin/bugreport.cgi?bug=1147093>

Docker build best practices

<https://docs.docker.com/build/building/best-practices/>

Attribution and use

bWAPP was created by Malik Mesellem. The installation source here is Larisa Moroz's Docker repository. The guide and helper files do not include the upstream application source or change its licence.

This is a dated record for a legacy training setup. Image availability and repository locations may change. Recheck the current source if a later installation produces a different error.